

Finite-Horizon Robust Suboptimal Control-Based Impact Angle Guidance

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This paper presents impact angle guidance based on finite-horizon robust optimal control. This approach, a fusion of newly proposed finite-time state-dependent Riccati equation and integral sliding-mode-control, solves finite-horizon tracking problem for input-affine nonlinear systems with specified terminal conditions. It also ensures robustness against external disturbances and uncertainties from beginning. The technique for ensuring specified final-time involves deriving closed-form control expression in terms of approximate solution of differential Riccati equation, which further reduces to solutions of algebraic Riccati and Lyapunov equations. This yields computationally efficient algorithm. To alleviate chattering, the supertwisting algorithm is incorporated with the proposed technique. Guidance strategy is derived after converting impact angle problem to tracking one using the proposed algorithm. Efficacy of the proposed guidance is vindicated for various engagement geometries, and also compared with an existing guidance strategy.

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I. INTRODUCTION

Guidance strategy plays a key role in ensuring not only zero miss distance but also to satisfy terminal constraints, such as impact angle, is important to increase the warhead effectiveness, the penetration capability, and reducing collateral damages. For practical situations, it is beneficial to achieve the desired impact angle within a finite time.

In literature, there exist a sliding mode control (SMC) based guidance laws for aligning interceptors to a desired direction within a finite time, and ensuring robustness against external disturbances and uncertainties [1]–[4]. In [1] and [2], the guidance laws were proposed based on the terminal SMC (TSMC) and finite-time convergence stability, respectively. In [3] and [4], the guidance laws were derived using the nonsingular TSMC, avoiding possible singularities during implementation. Although the guidance laws in [1]–[4] ensure finite time convergence, but the time of convergence was not fixed. Finite-time nature of engagement necessitates an efficient finite-time convergent guidance strategy to complete the mission in an optimal sense. In [5], the guidance law based on the integral SMC (ISM) and line-of-sight (LOS) shaping was proposed, which guarantees convergence of error to zero at interception. In [6], SMC-based guidance with look angle constraints was proposed. A finite time convergent guidance using the backstepping was proposed in [7].

It is well known that the optimal control approaches are perhaps well suited for reducing operation cost, structural loads, minimizing control effort, and achieving desired terminal conditions simultaneously [8]–[11]. This fact provided a thrust to design such guidance [12]–[17]. However, they require time-to-go estimates during implementation, which is a formidable challenge.

Moreover, an optimal control based trajectory optimization formulation often leads to a two-point boundary value problem, leading to the numerical iterative methods. These methods may not be feasible for real-time implementation. To address this issue, the approximate dynamic programming, followed by adaptive critic approach, is a real-time design technique, where such problem is solved using two neural networks [18], [19]. A potential drawback of this approach is that the domain in which state trajectory is likely to lie has to be known for training of networks. This is rather difficult to predict prior to the control design. Another promising approach is the linear quadratic regulator [20] or state-dependent Riccati equation (SDRE) [21], [22], to solve infinite-horizon optimal control problems. Typically, the guidance design demands a finite-horizon problem formulation [23], and hence, these methods may be not suitable for optimal guidance design with stringent final conditions. Recently, a finite-horizon SDRE (finite-SDRE) approach is proposed in [24]–[26]. However, this approach in [24]–[26] suffers from singularity problem of a matrix inverse, when the solution of differential Riccati equation (DRE) converges to the solution of algebraic Riccati equation (ARE). To avoid this problem, a solution approach, which needs to be heuristically selected, has been suggested, which may

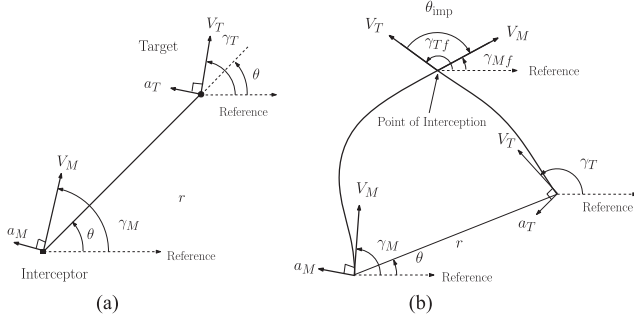


Fig. 1. Planar interceptor-target engagement. (a) Engagement geometry. (b) Impact angle.

not be effective for real-life practical problems, especially in missile guidance problem.

This motivates us to develop finite-time SDRE (FT-SDRE) approach for solving the problem of finite-horizon optimal tracking control for input-affine nonlinear systems with specified terminal conditions, while addressing above-mentioned constraints. The proposed approach is based on a modification in the SDRE method [21], [22] using the solution methodology of finite-horizon optimal regulator problem for a linear system [27], [28]. A brief discussion of this idea was presented in [29]. Its philosophy involves deriving a closed-form solution, in terms of an approximate solution of the matrix DRE for a specified final time. The solution can be further reduced to solutions of ARE and Lyapunov equation, and the state transition matrix of corresponding dynamic system. Thus, it becomes a computationally efficient method suitable for use in real-time applications.

Based on a fusion of the proposed FT-SDRE approach and ISMC method [30], a new nonlinear robust optimal control approach, called FT-SDRE-based ISMC, with fixed time error convergence, is presented in this paper to suppress the external disturbances and uncertainties as well. Moreover, to alleviate the chattering in SMC, the super-twisting (ST) algorithm [31]–[33] is also incorporated. Using this proposed method, a finite and fixed-time convergent impact angle guidance, which does not require time-to-go estimates, is also derived after converting impact angle problem to a tracking problem. Guidance strategy is designed to intercept stationary, constant-velocity and maneuvering targets at any impact angle, as well as to achieve the desired impact angle, at a fixed (pre-specified) final time. The proposed guidance strategy provides robustness against uncertainties (target maneuver) from beginning of engagement. Finally, in the simulation study, promising results indicate that this guidance design is quite successful and outperforms an existing guidance strategy in [3].

II. PROBLEM STATEMENT

Consider a planar interceptor-target engagement as shown in Fig. 1(a). The interceptor and the target are assumed to have constant speeds V_M and V_T , with their ratio $\nu = V_T/V_M < 1$, their flight path angles and lateral

accelerations are denoted by the pairs (γ_M, a_M) and (γ_T, a_T) , respectively. The relative distance and the LOS angle between target and interceptor are r and θ , respectively. The lateral accelerations of interceptor and target are denoted by a_M and a_T , respectively. The kinematic engagement equations are given by

$$\dot{r} = V_T \cos \theta_T - V_M \cos \theta_M \quad (1a)$$

$$r\dot{\theta} = V_T \sin \theta_T - V_M \sin \theta_M \quad (1b)$$

$$\dot{\gamma}_M = a_M/V_M, \quad \dot{\gamma}_T = a_T/V_T, \quad |a_T| \leq a_{T \max} \quad (1c)$$

where $\theta_T = \gamma_T - \theta$ and $\theta_M = \gamma_M - \theta$.

The objective of this paper is to design a guidance law, which enables the missile to intercept the target with a pre-specified angle within a finite time. In addition, the convergence of the missile to the desired impact direction should be at a fixed time. The impact angle, as shown in Fig. 1(b), denoted by θ_{imp} , is defined as the angle between the velocity vectors of the interceptor and the target at the time of interception, and is given by $\theta_{\text{imp}} \triangleq \gamma_{T_f} - \gamma_{M_f}$ where γ_{T_f} and γ_{M_f} are the final flight path angles of the target and the interceptor, respectively. Impact angle can be redefined in terms of final LOS-angle [1], and is given by

$$\theta_F = \gamma_{T_f} - \tan^{-1} \left(\frac{\sin \theta_{\text{imp}}}{\cos \theta_{\text{imp}} - \nu} \right). \quad (2)$$

Note that this relation was derived for nonmaneuvering targets. For stationary targets, the impact angle can be defined with respect to any reference (say $\gamma_{T_f} = 0^\circ$).

To ensure the objective of interception of the target at a prespecified impact angle, the guidance law needs to steer the interceptor on a collision course with desired final LOS angle at a specified (fixed) finite time.

The dynamics of LOS angle, obtained using (1)

$$\ddot{\theta} = -\frac{2\dot{r}\dot{\theta}}{r} - \left(\frac{\cos \theta_M}{r} \right) a_M + \left(\frac{\cos \theta_T}{r} \right) a_T \quad (3)$$

has relative degree 2 with respect to missile lateral acceleration a_M . It was shown in [1] that the $|\theta_M| = \pi/2$ is not an equilibrium point and hence, the lateral acceleration can be used to control LOS angle. Note that, the term with target acceleration can be treated as bounded uncertainty for the guidance design, due to unavailability of information on the target's maneuvers.

III. INTEGRAL SMC

To outline the ISMC algorithm [30], consider a control-affine nonlinear dynamical system represented by

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \mathbf{B}(\mathbf{x})\mathbf{u} + \mathbf{h}(\mathbf{x}, t) \quad (4)$$

where $\mathbf{x} \in \mathbb{R}^n$ is the state vector, $\mathbf{u} \in \mathbb{R}^m$ is the input vector, such that $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $\mathbf{B} : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$, with $\mathbf{B}(\mathbf{x}) \neq 0$ and $\text{rank}[\mathbf{B}(\mathbf{x})] = m \quad \forall \mathbf{x}$. The term $\mathbf{h}(\mathbf{x}, t)$ consists of external disturbances, parameter variations, and unmodeled dynamics, and is assumed to be bounded and of matched type. Suppose there exists a controller $\mathbf{u}_{\text{nom}}$, which gives the desired behavior of the states of nominal system (system without disturbances/uncertainties, that is, $\mathbf{h}(\mathbf{x}, t)$).

To design control $\mathbf{u}$ for the actual system (4) ensuring performance similar to a nominal system, the sliding manifold can be defined as $\sigma \triangleq \sigma_0(\mathbf{x}) + \sigma_I$, where $\sigma_0(\mathbf{x})$ consists of a linear combination of the states and σ_I is an integral term given by

$$\dot{\sigma}_I = -\frac{\partial \sigma_0}{\partial \mathbf{x}} [\mathbf{f}(\mathbf{x}) + \mathbf{B}(\mathbf{x})\mathbf{u}_{\text{nom}}], \sigma_I(0) = -\sigma_0(\mathbf{x}(0)). \quad (5)$$

The control input $\mathbf{u}$ is chosen as $\mathbf{u} = \mathbf{u}_{\text{nom}} + \mathbf{u}_{\text{disc}}$, where $\mathbf{u}_{\text{nom}}$ is the control input designed for nominal system and $\mathbf{u}_{\text{disc}}$ is the discontinuous control component designed to suppress the effect of unknown term $\mathbf{h}(\mathbf{x}, t)$. The selection of these components for the proposed approach in this paper will be discussed in Section V.

IV. FT-SDRE FOR FINITE-HORIZON NONLINEAR OPTIMAL CONTROL PROBLEM

In this section, the FT-SDRE approach for finite-horizon nonlinear optimal control problem is presented, and then used to design $\mathbf{u}_{\text{nom}}$ later.

A. Problem Formulation

Consider the nominal system described as

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \mathbf{B}(\mathbf{x})\mathbf{u}. \quad (6)$$

Assumption 1 $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $\mathbf{B} : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ are to be smooth, thus, continuous, in their domain.

The objective of the problem in this section is to design the optimal controller of the system in (6) which ensures tracking of system states at a fixed (specified) finite-time t_f , that is, $\mathbf{x}(t) \rightarrow \mathbf{x}_d(t)$ as $t \rightarrow t_f$, where $\mathbf{x}_d(t)$ is the arbitrarily varying bounded desired states. Note that the time t_f corresponds to the time at which states converge to their desired values, rather than the interception time or time of flight. The controller design will be based on a new FT-SDRE approach. In order to apply this approach, the system given by (6) can be expressed in terms of state-dependent co-efficient form

$$\dot{\mathbf{x}} = \mathbf{A}(\mathbf{x})\mathbf{x} + \mathbf{B}(\mathbf{x})\mathbf{u} \quad (7)$$

where $\mathbf{f}(\mathbf{x}) = \mathbf{A}(\mathbf{x})\mathbf{x}$, such that $\mathbf{A} : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times n}$.

The SDRE technique [21], [22] is popular for the infinite-horizon regulator problem. However, the aim here is to solve the finite-horizon nonlinear optimal control problem to track the arbitrarily varying desired states $\mathbf{x}_d$. Thus, to convert the tracking problem in the form of the regulator one, the error can be defined as $\mathbf{e} \triangleq \mathbf{x} - \mathbf{x}_d$ and its dynamics can be expressed as

$$\dot{\mathbf{e}} = \mathbf{f}(\mathbf{e} + \mathbf{x}_d) + \mathbf{B}(\mathbf{e} + \mathbf{x}_d)\mathbf{u} - \dot{\mathbf{x}}_d = \mathbf{f}_T(\mathbf{e}) + \mathbf{B}_T(\mathbf{e})\mathbf{u} \quad (8)$$

where $\mathbf{f}_T(\mathbf{e}) \triangleq \mathbf{f}(\mathbf{e} + \mathbf{x}_d) - \dot{\mathbf{x}}_d$ and $\mathbf{B}_T(\mathbf{e}) \triangleq \mathbf{B}(\mathbf{e} + \mathbf{x}_d)$. To factorize $\mathbf{f}_T(\mathbf{e})$ term into the form of $\mathbf{A}_T(\mathbf{e})\mathbf{e}$ for further use in the SDRE method, the condition $\mathbf{f}_T(0) = \mathbf{0}$ needs to be satisfied. But in this case, there might be a state-dependent bias term, say $\mathbf{g}(\mathbf{e})$, which makes $\mathbf{f}_T(0) \neq \mathbf{0}$,

that is

$$\mathbf{f}_T(\mathbf{e}) = \mathbf{A}_T(\mathbf{e})\mathbf{e} + \mathbf{g}(\mathbf{e}), \quad \mathbf{g}(0) \neq \mathbf{0}. \quad (9)$$

To solve this problem, the bias term $\mathbf{g}(\mathbf{e})$ can be expressed as

$$\mathbf{g}(\mathbf{e}) = [\mathbf{g}(\mathbf{e})/z]\mathbf{z} \quad (10)$$

where $\mathbf{z}$ is the slow varying auxiliary state with the stable dynamics, which is given by $\dot{\mathbf{z}} = -\eta\mathbf{z}$, $\eta > 0$, $\mathbf{z}(0) \neq \mathbf{0}$, where η is a small positive constant so that $\mathbf{g}(\mathbf{e})/z$ is smooth during the control period. The augmented system with states $\mathbf{e}_a = [\mathbf{e}^T \mathbf{z}^T]^T$ can be written as

$$\dot{\mathbf{e}}_a = \mathbf{A}_a(\mathbf{e}_a)\mathbf{e}_a + \mathbf{B}_a(\mathbf{e}_a)\mathbf{u} \quad (11a)$$

$$\mathbf{e}_{y_a} = \mathbf{e} = \mathbf{C}_a\mathbf{e}_a, \quad \mathbf{C}_a = [\mathbf{I}_n \quad \mathbf{0}] \quad (11b)$$

where

$$\mathbf{A}_a(\mathbf{e}_a) = \begin{bmatrix} \mathbf{A}_T(\mathbf{e}) & \mathbf{g}(\mathbf{e})/z \\ \mathbf{0} & -\eta \end{bmatrix}, \mathbf{B}_a(\mathbf{e}_a) = \begin{bmatrix} \mathbf{B}_T(\mathbf{e}) \\ \mathbf{0} \end{bmatrix}. \quad (12)$$

In order to use the infinite-horizon SDRE method for the system in (11a) and (11b), the cost function is given by

$$J_I = \frac{1}{2} \int_{t_0}^{\infty} [\mathbf{e}_{y_a}^T(\tau) \mathbf{Q}(\mathbf{e}_{y_a})\mathbf{e}_{y_a}(\tau) + \mathbf{u}^T(\tau) \mathbf{R}(\mathbf{e}_{y_a})\mathbf{u}(\tau)] d\tau \quad (13)$$

where the error and input weighting matrices (design parameters) satisfy $\mathbf{Q}(\mathbf{e}_{y_a}) \geq 0$ and $\mathbf{R}(\mathbf{e}_{y_a}) > 0$ for all $\mathbf{e}_a$, respectively. For simplicity in the notations, the arguments of the vectors and scalars are omitted at some places in rest of the paper.

Assumption 2 The pair $\{\mathbf{A}_a, \mathbf{B}_a\}$ and $\{\mathbf{A}_a, \mathbf{Q}_a^{1/2}\}$ are pointwise stabilizable and detectable parameterization, respectively, of the system given by (11a) and (11b), for all $\mathbf{e}_a$, where $\mathbf{Q}_a = \mathbf{C}_a^T \mathbf{Q} \mathbf{C}_a$.

The controller for this infinite-time horizon optimal control problem can be obtained as

$$\mathbf{u}(\mathbf{e}_a) = -\mathbf{R}^{-1} \mathbf{B}_a^T \mathbf{P}_{ss}(\mathbf{e}_a) \mathbf{e}_a \quad (14)$$

with matrix $\mathbf{P}_{ss}(\mathbf{e}_a) := \mathbf{P}_{ss}$ computed using the ARE [21], [22], with $\mathbf{M} = \mathbf{B}_a \mathbf{R}^{-1} \mathbf{B}_a^T$ as

$$\mathbf{A}_a^T \mathbf{P}_{ss} + \mathbf{P}_{ss} \mathbf{A}_a - \mathbf{P}_{ss} \mathbf{M} \mathbf{P}_{ss} + \mathbf{C}_a^T \mathbf{Q} \mathbf{C}_a = 0. \quad (15)$$

Now, to solve the finite-time horizon nonlinear optimal control problem using an FT-SDRE approach, the cost function for the system in (11a) and (11b) is chosen as

$$J_F = \frac{1}{2} \mathbf{e}_{y_a}^T(t_f) \mathbf{H}_f \mathbf{e}_{y_a}(t_f) + \frac{1}{2} \int_{t_0}^{t_f} [\mathbf{e}_{y_a}^T(\tau) \mathbf{Q}(\mathbf{e}_a) \mathbf{e}_{y_a}(\tau) + \mathbf{u}^T(\tau) \mathbf{R}(\mathbf{e}_a) \mathbf{u}(\tau)] d\tau \quad (16)$$

where the error weighting matrix (design parameter) at time instant t_f satisfies $\mathbf{H}_f \geq 0$. The corresponding finite horizon controller is computed by the following expression:

$$\mathbf{u}(\mathbf{e}_a, t) = -\mathbf{R}^{-1} \mathbf{B}_a^T \mathbf{P}(\mathbf{e}_a, t) \mathbf{e}_a \quad (17)$$

where $\mathbf{P}(\mathbf{e}_a, t) := \mathbf{P}(t)$ is a solution to DRE

$$-\dot{\mathbf{P}} = \mathbf{A}_a^T \mathbf{P} + \mathbf{P} \mathbf{A}_a - \mathbf{P} \mathbf{M} \mathbf{P} + \mathbf{C}_a^T \mathbf{Q} \mathbf{C}_a \quad (18)$$

with the boundary condition is given by

$$\mathbf{P}(t_f) = \mathbf{C}_a^T \mathbf{H}_f \mathbf{C}_a. \quad (19)$$

To satisfy the boundary condition, (19), the infinite-horizon SDRE method is not applicable here. Also, the solution of finite-horizon problem is dependent on the solution of DRE, (18), rather than ARE, (15). The solution of DRE, (18), can be obtained using the backward integration with the boundary condition, (19). But the states at future time is not known ahead, so one cannot compute such solution.

A remedy of this problem based on approximate analytical approach is proposed here.

REMARK 1 The relation between an approximate control solution, given by (17) using the solution of DRE in (18), and the exact optimal solution to the problem investigated in [25].

B. Solving Finite-Horizon Nonlinear Optimal Control Problem Using FT-SDRE

In this section, the approach for solving (18) in terms of solutions of ARE and Lyapunov algebraic equation using the Hamiltonian method [27], [28] is discussed. The underlying concept for this proposed approach is that the state values in the Riccati equation, which are computed at their current values at each time step t , are assumed to be constant from t to t_f . Based on this assumption, an approximate closed-form solution will be derived, by using a transformation with the solution of ARE, which converts the original nonlinear Riccati equation to a Lyapunov equation. Subtracting (15) from (18) leads to

$$-\dot{\mathbf{P}}(t) = \mathbf{A}_a^T [\mathbf{P}(t) - \mathbf{P}_{ss}] + [\mathbf{P}(t) - \mathbf{P}_{ss}] \mathbf{A}_a - \mathbf{P}(t) \mathbf{M} \mathbf{P}(t) + \mathbf{P}_{ss} \mathbf{M} \mathbf{P}_{ss} \quad (20)$$

where $\mathbf{P}_{ss}$ is the steady-state value of $\mathbf{P}(t)$ and can be obtained from (15), which is the corresponding ARE for (18). Now, the error between the Riccati matrices $\mathbf{P}(t)$ and $\mathbf{P}_{ss}$ is defined by $\mathbf{P}_e(t) \triangleq \mathbf{P}(t) - \mathbf{P}_{ss}$. Substituting $\mathbf{P}_e(t)$ in (20) produces

$$-\dot{\mathbf{P}}_e(t) = \mathbf{A}_a^T \mathbf{P}_e(t) + \mathbf{P}_e(t) \mathbf{A}_a - \mathbf{P}(t) \mathbf{M} \mathbf{P}(t) + \mathbf{P}_{ss} \mathbf{M} \mathbf{P}_{ss}. \quad (21)$$

After simplifying RHS of (21), we get

$$-\dot{\mathbf{P}}_e(t) = \mathbf{A}_r^T \mathbf{P}_e(t) + \mathbf{P}_e(t) \mathbf{A}_r - \mathbf{P}_e(t) \mathbf{M} \mathbf{P}_e(t) \quad (22)$$

where the matrix $\mathbf{A}_r$ (stable) and boundary condition are

$$\mathbf{A}_r \triangleq [\mathbf{A}_a - \mathbf{M} \mathbf{P}_{ss}], \mathbf{P}_e(t_f) \triangleq [\mathbf{P}(t_f) - \mathbf{P}_{ss}]. \quad (23)$$

According to the Hamiltonian method, the solution of the differential equation (22) can be represented in the form

$$\mathbf{P}_e(t) = \mathbf{L}(t) \mathbf{S}^{-1}(t), \mathbf{L}(t) := \mathbf{L}(\mathbf{e}_a, t), \mathbf{S}(t) := \mathbf{S}(\mathbf{e}_a, t). \quad (24)$$

The time derivative of (24) gives

$$-\dot{\mathbf{P}}_e(t) = [-\dot{\mathbf{L}}(t) + \mathbf{L}(t) \mathbf{S}^{-1}(t) \dot{\mathbf{S}}(t)] \mathbf{S}^{-1}(t). \quad (25)$$

Equation (22) can be expressed using $\mathbf{L}(t)$ and $\mathbf{S}(t)$ as

$$-\dot{\mathbf{P}}_e(t) = [\mathbf{A}_r^T \mathbf{L}(t) + \mathbf{L}(t) \mathbf{S}^{-1} \{\mathbf{A}_r \mathbf{S}(t) - \mathbf{M} \mathbf{L}(t)\}] \mathbf{S}^{-1}(t). \quad (26)$$

On comparing (25) with (26) results in

$$\begin{bmatrix} \dot{\mathbf{S}}(t) \\ \dot{\mathbf{L}}(t) \end{bmatrix} = \begin{bmatrix} \mathbf{A}_r & -\mathbf{M} \\ 0 & -\mathbf{A}_r^T \end{bmatrix} \begin{bmatrix} \mathbf{S}(t) \\ \mathbf{L}(t) \end{bmatrix} \quad (27)$$

with the boundary conditions given by

$$[\mathbf{S}(t_f) \ \mathbf{L}(t_f)]^T = [\mathbf{I} \ \mathbf{P}(t_f) - \mathbf{P}_{ss}]^T. \quad (28)$$

Equation (27) can be block diagonalized using the transformation

$$\mathbf{N}(t) = \mathbf{S}(t) - \mathbf{H}_{ss}(\mathbf{e}_a) \mathbf{L}(t) \quad (29)$$

where $\mathbf{H}_{ss}(\mathbf{e}_a) := \mathbf{H}_{ss}$ is obtained using the Lyapunov algebraic equation given by

$$\mathbf{A}_{cl} \mathbf{H}_{ss} + \mathbf{H}_{ss} \mathbf{A}_{cl}^T - \mathbf{M} = 0. \quad (30)$$

Now, using (27) and (30) in the time derivative of (29), the following can be obtained:

$$\dot{\mathbf{N}}(t) = \mathbf{A}_r [\mathbf{S}(t) - \mathbf{H}_{ss} \mathbf{L}(t)] = \mathbf{A}_r \mathbf{N}(t) \quad (31)$$

with the corresponding boundary condition given by

$$\mathbf{N}(t_f) = \mathbf{S}(t_f) - \mathbf{H}_{ss} \mathbf{L}(t_f) = \mathbf{I} - \mathbf{H}_{ss} \mathbf{L}(t_f). \quad (32)$$

The solution of $\mathbf{N}(t)$ from (31) is obtained using (32)

$$\mathbf{N}(t) = e^{[t-t_f] \mathbf{A}_r} \mathbf{N}(t_f) = e^{-\Gamma} [\mathbf{I} - \mathbf{H}_{ss} \mathbf{L}(t_f)] \quad (33)$$

where Γ is defined as $\Gamma \triangleq [t_f - t] \mathbf{A}_r$. Using (28), the solution of $\mathbf{L}(t)$ from (27) is computed as

$$\mathbf{L}(t) = e^{-[t-t_f] \mathbf{A}_r^T} \mathbf{L}(t_f) = e^{\Gamma^T} \mathbf{L}(t_f). \quad (34)$$

The expression of $\mathbf{S}(t)$ using (29), (33), and (34)

$$\begin{aligned} \mathbf{S}(t) &= \mathbf{N}(t) + \mathbf{H}_{ss} \mathbf{L}(t) = e^{-\Gamma} [\mathbf{I} - \mathbf{H}_{ss} \mathbf{L}(t_f)] \\ &\quad + \mathbf{H}_{ss} e^{\Gamma^T} \mathbf{L}(t_f) \\ &= \Phi_S(t - t_f) \mathbf{S}(t_f) = \Phi_S(t - t_f) \mathbf{I} \end{aligned} \quad (35)$$

where $\Phi_S(t - t_f)$ is the state transition matrix, which is nonsingular for all $t \in [t_0, t_f]$ [27], based on Assumption 2. This implies that $\mathbf{S}(t)$ is a nonsingular matrix for all $t \in [t_0, t_f]$. Moreover, $\mathbf{L}(t)$ in (34) exists for all $t \in [t_0, t_f]$. Using (24), (34), and (35)

$$\begin{aligned} \mathbf{P}_e(t) &= e^{\Gamma^T} \mathbf{L}(t_f) \left[e^{-\Gamma} [\mathbf{I} - \mathbf{H}_{ss} \mathbf{L}(t_f)] + \mathbf{H}_{ss} e^{\Gamma^T} \mathbf{L}(t_f) \right]^{-1} \\ &= e^{\Gamma^T} \mathbf{L}(t_f) \left[\{\mathbf{I} - \mathbf{H}_{ss} \mathbf{L}(t_f)\} + e^{\Gamma} \mathbf{H}_{ss} e^{\Gamma^T} \mathbf{L}(t_f) \right]^{-1} e^{\Gamma} \end{aligned} \quad (36)$$

exists for all $t \in [t_0, t_f]$. A closed-form expression for $\mathbf{P}(t)$ can be written using (23)

$$\mathbf{P}(t) = \mathbf{P}_{ss} + \mathbf{P}_e(t) \quad (37)$$

where $\mathbf{P}_e(t)$ can be computed by (36). Note that the controller (17) of the FT-SDRE approach uses this expression for $\mathbf{P}(t)$ in (37), which is an approximate closed-form solution of DRE (18). Hence, the proposed FT-SDRE approach results into a suboptimal closed-loop controller.

REMARK 2 Unlike the approaches in [24]–[26], the proposed FT-SDRE approach here is not required to have any heuristical design selection to avoid this matrix inversion. Thus, it can be effective for real-life practical problems, especially in a missile guidance problem.

REMARK 3 The selection of the weighting matrices $\mathbf{Q}(\mathbf{e}_a)$, $\mathbf{R}(\mathbf{e}_a)$, and $\mathbf{H}_f$ is a tradeoff between control effort $\mathbf{u}(t)$, output error $\mathbf{e}_{y_a}(t)$, and terminal output error $\mathbf{e}_{y_a}(t_f)$. For missile guidance design, $\mathbf{H}_f$ is usually chosen higher than $\mathbf{Q}(\mathbf{e}_a)$, so that the desired terminal constraints can be achieved as close as possible. For more details on selecting these weighting matrices, one can refer to [22].

C. Stability Analysis of FT-SDRE

The stability of the output ($\mathbf{e}_{y_a} = \mathbf{e}$) given by (11b) of the system (11a) using the FT-SDRE will be analyzed based on [34] in this section.

THEOREM 1 FT-SDRE approach renders local exponential stability of the output (11b) of the system (11a).

PROOF Consider the matrix $\mathbf{A}_{cl} = \mathbf{A}_a - \mathbf{M}\mathbf{P}$, given as

$$\mathbf{A}_{cl}(\mathbf{e}, z, t) = \mathbf{A}_{cl_0}(z, t) + \mathbf{D}(\mathbf{e}, z, t) \quad (38)$$

where $\mathbf{A}_{cl_0}(z, t) = \lim_{\|\mathbf{e}\| \rightarrow 0} \mathbf{A}_{cl}(\mathbf{e}, z, t)$, and $\mathbf{D} : \mathbb{R}^n \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}^{(n+1) \times (n+1)}$ is a matrix-valued function, which is continuous on $\mathbf{e}$ and z , represented by

$$\lim_{\|\mathbf{e}\| \rightarrow 0} \mathbf{D}(\mathbf{e}, z, t) = \mathbf{0} \quad \forall t \in [t_0, t_f]. \quad (39)$$

The matrix $\mathbf{A}_{cl_0}(z, t)$ can also be described as

$$\mathbf{A}_{cl_0}(z, t) = \mathbf{A}_{a_0} - \mathbf{M}_0 \mathbf{P}_0(t) = \mathbf{A}_{a_0} - \mathbf{M}_0 \mathbf{P}_0(t) \quad (40)$$

where $\mathbf{A}_{a_0} \equiv \mathbf{A}_a(z)$, $\mathbf{B}_{a_0} \equiv \mathbf{B}_a(0)$, $\mathbf{Q}_{a_0} \equiv \mathbf{Q}_a(0)$, $\mathbf{R}_0 \equiv \mathbf{R}(0)$, $\mathbf{M}_0 = \mathbf{B}_{a_0} \mathbf{R}_0^{-1} \mathbf{B}_{a_0}^T$, and $\mathbf{P}_0$ is the solution to the DRE

$$-\dot{\mathbf{P}}_0(t) = \mathbf{P}_0(t) \mathbf{A}_{a_0} + \mathbf{A}_{a_0}^T \mathbf{P}_0(t) - \mathbf{P}_0(t) \mathbf{M}_0 \mathbf{P}_0(t) + \mathbf{Q}_{a_0} \quad (41)$$

with $\mathbf{P}_0(t_f) = \mathbf{P}(t_f)$. The following can be written as

$$\lim_{\|\mathbf{e}\| \rightarrow 0} \mathbf{P}(\mathbf{e}, z, t) = \mathbf{P}(0, z, t) \equiv \mathbf{P}_0(t) \quad \forall t \in [t_0, t_f]. \quad (42)$$

For the proofs of (40) and (42), one may refer to Lemmas 2 and 3, respectively, in [34].

Let $V_1(\mathbf{e}, t) = \mathbf{e}^T \mathbf{P}_e \mathbf{e} = \mathbf{e}^T \mathbf{C}_a \mathbf{P}_0 \mathbf{C}_a^T \mathbf{e}$, where $\mathbf{P}_e = \mathbf{C}_a \mathbf{P}_0 \mathbf{C}_a^T$, be a Lyapunov function, which satisfies

$$\rho_1 \|\mathbf{e}\|^2 \leq V_1(\mathbf{e}, t) \leq \rho_2 \|\mathbf{e}\|^2 \quad \forall \mathbf{e} \in \Omega \quad \forall t \in [t_0, t_f] \quad (43)$$

where Ω is a bounded subset of $\mathbb{R}^{n+1}$ and

$$\begin{aligned} \rho_1 &\equiv \min_{i \in \{1, \dots, n\}, t \in [t_0, t_f]} \lambda_i(\mathbf{C}_a \mathbf{P}_0 \mathbf{C}_a^T) \\ \rho_2 &\equiv \max_{i \in \{1, \dots, n\}, t \in [t_0, t_f]} \lambda_i(\mathbf{C}_a \mathbf{P}_0 \mathbf{C}_a^T). \end{aligned} \quad (44)$$

Using (17) and (38), the closed-loop dynamics of augmented system, given by (11a) and (11b), can be written as

$$\dot{\mathbf{e}} = \mathbf{C}_a \dot{\mathbf{e}}_a = \mathbf{C}_a \mathbf{A}_{cl}(\mathbf{e}, z, t) \mathbf{e}_a = \mathbf{C}_a [\mathbf{A}_{cl_0}(z, t) + \mathbf{D}(\mathbf{e}, z, t)] \mathbf{e}_a. \quad (45)$$

Taking the total derivative of V_1 , which is derived along the closed-loop trajectory, leads to

$$\begin{aligned} \dot{V}_1 &= \dot{\mathbf{e}}^T \mathbf{C}_a \mathbf{P}_0(t) \mathbf{C}_a^T \mathbf{e} + \mathbf{e}^T \mathbf{C}_a \dot{\mathbf{P}}_0(t) \mathbf{C}_a^T \mathbf{e} + \mathbf{e}^T \mathbf{C}_a \mathbf{P}_0(t) \mathbf{C}_a^T \dot{\mathbf{e}} \\ &= \mathbf{e}_a^T \left[-\mathbf{C}_a^T \mathbf{C}_a \mathbf{P}_0(t) \mathbf{C}_a^T \mathbf{C}_a \mathbf{B}_{a_0} \mathbf{R}_0^{-1} \mathbf{B}_{a_0}^T \mathbf{P}_0(t) \right. \\ &\quad \left. - \mathbf{C}_a^T \mathbf{C}_a \mathbf{Q}_{a_0} \mathbf{C}_a^T \mathbf{C}_a + \Lambda(\mathbf{e}, z, t) \right] \mathbf{e}_a \end{aligned} \quad (46)$$

where $\Lambda(\mathbf{e}, z, t) \equiv \mathbf{C}_a^T \mathbf{C}_a \mathbf{P}_0(t) \mathbf{C}_a^T \mathbf{C}_a \mathbf{D}(\mathbf{e}, z, t) + \mathbf{D}^T(\mathbf{e}, z, t) \mathbf{C}_a^T \mathbf{C}_a \mathbf{P}_0(t) \mathbf{C}_a^T \mathbf{C}_a$. Using (11b), (12), and after some algebraic manipulations, (46) can be written as

$$\dot{V}_1 = \mathbf{e}^T \left[-\mathbf{P}_{011}(t) \mathbf{B}_{T_0} \mathbf{R}_0^{-1} \mathbf{B}_{T_0}^T \mathbf{P}_{011}(t) - \mathbf{Q}_0 + \Lambda_{11}(\mathbf{e}, z, t) \right] \mathbf{e} \quad (47)$$

where $\mathbf{P}_0 = \begin{bmatrix} [\mathbf{P}_{011}]_{n \times n} & [\mathbf{P}_{012}]_{n \times 1} \\ [\mathbf{P}_{021}]_{1 \times n} & [\mathbf{P}_{022}]_{1 \times 1} \end{bmatrix}$, $\Lambda = \begin{bmatrix} [\Lambda_{11}]_{n \times n} & [\Lambda_{12}]_{n \times 1} \\ [\Lambda_{21}]_{1 \times n} & [\Lambda_{22}]_{1 \times 1} \end{bmatrix}$, $\mathbf{Q}_0 \equiv \mathbf{Q}(0)$, $\mathbf{B}_{T_0} \equiv \mathbf{B}_T(0)$.

Suppose, $\gamma(\mathbf{e}, z, t)$ is a real-valued function, which is the maximum of absolute value of eigenvalues of $\Lambda(\mathbf{e}, z, t)$ for given $\mathbf{e}$, z , and t . Using (39), we have

$$\lim_{\|\mathbf{e}\| \rightarrow 0} \gamma(\mathbf{e}, z, t) = 0 \quad \forall t \in [t_0, t_f]. \quad (48)$$

Now the following can be expressed as

$$\begin{aligned} &\max_{i \in \{1, \dots, n\}, \mathbf{e}_a \in \Pi, t \in [t_0, t_f]} \lambda_i \left[-\mathbf{P}_{011}(t) \mathbf{B}_{T_0} \mathbf{R}_0^{-1} \mathbf{B}_{T_0}^T \mathbf{P}_{011}(t) - \mathbf{Q}_0 \right. \\ &\quad \left. + \Lambda_{11}(\mathbf{e}, z, t) \right] \\ &\leq -\min_{i \in \{1, \dots, n\}, t \in [t_0, t_f]} \lambda_i \left[\mathbf{P}_{011}(t) \mathbf{B}_{T_0} \mathbf{R}_0^{-1} \mathbf{B}_{T_0}^T \mathbf{P}_{011}(t) + \mathbf{Q}_0 \right] \\ &\quad + \max_{i \in \{1, \dots, n\}, \mathbf{e}_a \in \Pi, t \in [t_0, t_f]} \lambda_i [\Lambda_{11}(\mathbf{e}, z, t)] \\ &\leq -\mu + \gamma(\mathbf{e}, z, t) \end{aligned} \quad (49)$$

where $\mu \triangleq \min_{i \in \{1, \dots, n\}, t \in [t_0, t_f]} \lambda_i [\mathbf{P}_{011}(t) \mathbf{B}_{T_0} \mathbf{R}_0^{-1} \mathbf{B}_{T_0}^T \mathbf{P}_{011}(t) + \mathbf{Q}_0]$. Based on Assumption 2, it can be concluded that μ is a positive and bounded quantity. Also, if $\mu > \gamma(\mathbf{e}, z, t)$, it can be observed from (48) that $\dot{V}_1 < 0$ is satisfied. By (48), it can be obtained that the set $\Pi = \{\mathbf{e} \in \mathbb{R}^n, z \in \mathbb{R} : \gamma(\mathbf{e}, z, t) < \mu, \forall t \in [t_0, t_f]\}$ is a nonempty set. As $\gamma(0, z, t) = 0 \quad \forall t, \mathbf{e} = \mathbf{0}$ is an interior point of Π . Furthermore, $\gamma(\mathbf{e}, z, t)$ is continuous on $\mathbf{e}$ and z . Thus, $\dot{V}_1 < 0$ for $\mathbf{e} \in \Pi$, which can be written as

$$\dot{V}_1(\mathbf{e}, t) \leq -\rho_3 \|\mathbf{e}\|^2, \quad \mathbf{e} \in \Pi, \quad t \in [t_0, t_f] \quad (50)$$

where

$$\begin{aligned} \rho_3 \equiv & \min_{i \in \{1, \dots, n\}, \{e, z\} \in \Pi, t \in [t_0, t_f]} \\ & \lambda_i [P_{011}(t) B_{T_0} R_0^{-1} B_{T_0}^T P_{011}(t) + Q_0 \\ & - \Lambda_{11}(e, z, t)] > 0. \end{aligned} \quad (51)$$

When $\Omega = \Pi$, (51) with (44) lead to local exponential stability of the output of the system. ■

V. OPTIMAL INTEGRAL SMC DESIGN BASED ON FT-SDRE

A new optimal sliding mode controller, based on ISMC and FT-SDRE technique, is proposed here. To do this, the system dynamics (4) can be represented in the form of (11a)

$$\dot{e}_a = A_a e_a + B_a(e_a) u + h_a(e_a, t) \quad (52)$$

where $h_a(e_a, t) = [h^T(x, t) \ 0]^T := h_a$. Also, the output equation for the system dynamics (52) is the same as (11b).

To design the controller for these system dynamics and output equation, define a switching surface as

$$\sigma = G \int_0^t [\dot{e}_{y_a}(\tau) - C_a \{A_a - B_a R^{-1} B_a^T P(\tau)\} e_a(\tau)] d\tau \quad (53)$$

where G is a design matrix, R and P are matrices of appropriate dimensions defined in (16) and (37), respectively. This switching surface can be rewritten, by using (11b)

$$\begin{aligned} \sigma = & G C_a [e_a(t) - e_a(0)] \\ & - G C_a \int_0^t [\{A_a - B_a R^{-1} B_a^T P(\tau)\} e_a(\tau)] d\tau. \end{aligned} \quad (54)$$

On differentiating (54), it leads to

$$\dot{\sigma} = G C_a \dot{e}_a - G C_a A_a e_a + G C_a B_a R^{-1} B_a^T P(t) e_a. \quad (55)$$

Substituting (52) in (55), $\dot{\sigma}$ reduces to

$$\dot{\sigma} = G C_a B_a u + G C_a h_a + G C_a B_a R^{-1} B_a^T P(t) e_a. \quad (56)$$

THEOREM 2 For a system given by (52) and (11b), if the switching surface is chosen as in (54) with controller

$$u = u_{\text{nom}} + u_{\text{disc}} \quad (57)$$

$$u_{\text{nom}} = -R^{-1} B_a^T P(t) e_a \quad (58a)$$

$$u_{\text{disc}} = -[G C_a B_a]^{-1} D_g \text{sign}(\sigma),$$

$$D_g = \text{diag}(d_{g_1}, \dots, d_{g_m})$$

$$\text{sign}(\sigma) = [\text{sign}(\sigma_1) \ \dots \ \text{sign}(\sigma_m)]^T \quad (58b)$$

the matrix $P(t)$ is obtained using (37), $D_g := D_g(e_a)$ is the discontinuous control gain matrix, with $|h_{g_i}| < d_{g_i} \forall i = 1, 2, \dots, m$, and G is selected such that $[G C_a B_a]$ is a full rank matrix, then the output of the system is locally exponentially stable.

PROOF The proof of this theorem can be performed using the Lyapunov stability theory, similar to those in [31]–[33].

Once $\dot{V}_\sigma < 0$ is satisfied, the sliding mode occurs in the system and results in convergence of σ to zero. When the system is in sliding mode, the control input reduces to that using FT-SDRE, which further ensures the convergence of error e_{y_a} in the vicinity of zero. ■

REMARK 4 In this proposed approach, σ is zero from initial time due to the particular choice of σ in (53) [30]. As a consequence, the reaching phase is completely eliminated and, thus, the robustness against uncertainty h_a is ensured from the beginning. Hence, this approach gives the optimal performance of this closed-loop system.

REMARK 5 For optimal selection of G , one can refer to [35].

The discontinuous control component u_{disc} in (58b) may lead to chattering, which is a well-known problem with sliding mode. To alleviate this chattering, the ST control algorithm [31]–[33] can be implemented as follows.

COROLLARY 1 Along with the equivalent control component u_{nom} in (58a) and the switching surface chosen in (54) for the same system mentioned in Theorem 1, if u_{disc} in (57) is considered as

$$u_{\text{disc}} = [G C_a B_a]^{-1} [-\alpha |\sigma|^\gamma \text{sign}(\sigma) + w] \quad (59)$$

where

$$|\sigma|^\gamma = \text{diag}(|\sigma_1|^\gamma, \dots, |\sigma_m|^\gamma), \dot{w} = -\beta \text{sign}(\sigma) \quad (60)$$

then the output of the system is locally exponentially stable.

PROOF Using (58a) and (59), we have

$$\dot{\sigma} = -\alpha |\sigma|^\gamma \text{sign}(\sigma) + w + h_g, \dot{w} = -\beta \text{sign}(\sigma). \quad (61)$$

With the appropriate choice of α , β , and γ , the stability of (61) can be easily proved [31]–[33]. ■

VI. GUIDANCE DESIGN USING FT-SDRE-BASED ISMC

In this section, the impact angle guidance law for the maneuvering targets will be derived, and the constant velocity and stationary targets can be considered as special cases of the same. For this design, the dynamics of LOS angle in (3) can be rewritten in the form of state-space representation as

$$\dot{x}_1 = x_2 \quad (62a)$$

$$\dot{x}_2 = -\frac{2\dot{r}x_2}{r} - \left[\frac{\cos(\gamma_M - x_1)}{r} \right] u + \left[\frac{\cos(\gamma_T - x_1)}{r} \right] a_T \quad (62b)$$

where $x_1 = \theta$ and $x_2 = \dot{\theta}$ are the system states, and $u = a_M$ is the control input. In a vector-matrix notational form, with system states $x = [x_1, x_2]^T$, the system can be described by

$$\dot{x} = A(x)x + B(x)u + h(x, t) \quad (63)$$

where

$$\begin{aligned} \mathbf{x} &= [x_1 \quad x_2]^T, \mathbf{A}(\mathbf{x}) = \begin{bmatrix} 0 & 1 \\ 0 & -2\dot{r}/r \end{bmatrix} \\ \mathbf{B}(\mathbf{x}) &= \begin{bmatrix} 0 & -\cos(\gamma_M - x_1)/r \end{bmatrix}^T \\ \mathbf{h}(\mathbf{x}, t) &= \begin{bmatrix} 0 & \cos(\gamma_T - x_1)/r \end{bmatrix}^T a_T. \end{aligned} \quad (64)$$

Note that, it is difficult to predict the exact target maneuver, and thus it is considered as unknown uncertainty. It is also assumed that the maximum maneuvering capacity of target, $a_T^{\max}$, is known, such that, $|a_T| \leq a_T^{\max}$. Objectives of a guidance design are to make LOS rate $\dot{\theta} = 0$ and to achieve the desired final LOS angle θ_F , given by (2), which is a time-varying quantity for maneuvering targets. This guidance law needs to satisfy the following, within a fixed final time

$$x_1 \rightarrow x_{d1} \{= \theta_F\}; x_2 \rightarrow x_{d2} \{= \dot{\theta}_F\}. \quad (65)$$

Therefore, this tracking problem in case of the maneuvering targets is required to convert into the form of a regulator problem for designing the proposed FT-SDRE-based ISMC guidance. Now, using (65), the system dynamics (63) and corresponding output equation can be represented, respectively, in the form of (52) and (11b)

$$\begin{aligned} \dot{\mathbf{e}}_a &= \mathbf{A}_a \mathbf{e}_a + \mathbf{B}_a u + \mathbf{h}_a, \quad \mathbf{e}_{y_a} = \mathbf{C}_a \mathbf{e}_a \quad (66) \\ \mathbf{e}_a &= \begin{bmatrix} x_1 - x_{d1} \\ x_2 - x_{d2} \\ z \end{bmatrix}, \mathbf{C}_a = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \\ \mathbf{A}_a &= \begin{bmatrix} 0 & 1 & (x_{d2} - \dot{x}_{d1})/z \\ 0 & -2\dot{r}/r & -(2\dot{r}x_{d2}/r + \dot{x}_{d2})/z \\ 0 & 0 & -\eta \end{bmatrix} \\ \mathbf{B}_a &= \begin{bmatrix} 0 & -\cos(\gamma_M - x_1)/r & 0 \end{bmatrix}^T \\ \mathbf{h}_a &= \begin{bmatrix} 0 & \cos(\gamma_T - x_1)/r & 0 \end{bmatrix}^T a_T. \end{aligned} \quad (67)$$

Thus, the guidance command against maneuvering targets, computed using proposed FT-SDRE-based ST ISMC approach, is given by

$$u = a_M = a_{M_{\text{nom}}} + a_{M_{\text{disc}}} \quad (68)$$

$$a_{M_{\text{nom}}} = -\mathbf{R}^{-1} \mathbf{B}_a^T \mathbf{P}(t) \mathbf{e}_a \quad (69a)$$

$$a_{M_{\text{disc}}} = [\mathbf{G} \mathbf{C}_a \mathbf{B}_a]^{-1} (-\alpha |\sigma|^\gamma \text{sign}(\sigma) + \mathbf{w}) \quad (69b)$$

$$\dot{\mathbf{w}} = -\beta \text{sign}(\sigma). \quad (69c)$$

Note that the nominal controller $a_{M_{\text{nom}}}$ is derived for the nominal system, which is given by $\dot{\mathbf{e}}_a = \mathbf{A}_a \mathbf{e}_a + \mathbf{B}_a u_{\text{nom}}$, where $\mathbf{P}(t)$ is computed given by (37).

REMARK 6 Even though the guidance law is derived for maneuvering targets, the cases of stationary and nonmaneuvering targets are treated as special cases.

REMARK 7 For the case of moving but nonmaneuvering targets, $a_T = 0$ and hence $\mathbf{h}(\mathbf{x}, t) = 0$. For stationary tar-

gets, target speed V_T is also zero along with $a_T = 0$. It is worth to mention here that, for both types of targets, the system dynamics remains the same as that of maneuvering targets, except the uncertainty term $\mathbf{h}(\mathbf{x}, t)$. Guidance laws for such targets also remain unchanged.

VII. SIMULATION STUDY

In this section, performance of the proposed guidance law is evaluated through numerical simulations. The lateral acceleration of the interceptor is assumed to be bounded with the maximum lateral acceleration $a_M^{\max}$. Based on extensive simulation trials, the matrices $\mathbf{R}$, $\mathbf{H}_f$, and $\mathbf{Q}$, and other guidance parameters are chosen as

$$\begin{aligned} \mathbf{Q} &= \text{diag}(10^5, 10^5); \mathbf{H}_f = \text{diag}(10^9, 10^{10}); \mathbf{R} = 1, \gamma = 0.5 \\ \alpha &= 1.5 \times 10^{-3}, \beta = 1.1 \times 10^{-3}, \eta = 10^{-5}, \mathbf{G} = [11]. \end{aligned}$$

The initial range and LOS angle are assumed to be 10 km and 0° , while the initial flight path angles of missile and target are 30° and 120° , respectively. The speeds of missile and target are 500 and 400 m/s, unless the target is assumed to be stationary, and $a_M^{\max}$ is 400 m/s². The start points of missile and target are denoted by diamond and circle markers, respectively, in the subsequent figures. The thin dotted lines in subsequent figures represent the desired reference values of corresponding variables.

Note that the LOS and its rate converge to their corresponding desired values at a prespecified time. However, the time of target interception may be larger than the specified fixed time. As $t > t_f$, it is clear from expression for Γ that it no longer remains a Hurwitz matrix and thus the matrix $\mathbf{P}_e$, (36), may encounter some issue during the implementation of the proposed method. Note that t_f corresponds to the time at which missile needs to be aligned in specific direction. To avoid such problems, the duration of final time has been considered with a sliding time with a finite horizon, ϵ . This essentially means that when $t_f - t < \epsilon$, then $t_{f_{\text{new}}} = t_{f_{\text{old}}} + \epsilon$, which makes the guidance commands remain applicable. The sliding of t_f continues till the missile intercepts the target. It helps to ensure the errors within their tolerance bounds, resulting in alignment of the interceptor in a desired impact direction. To ensure the target interception at a desired impact angle, t_f should not be chosen to be larger than time of interception.

A. Stationary Targets

A case of stationary target with a desired impact angle of 30° is first considered. The initial conditions and simulation parameters are same as mentioned above. To evaluate the ability of the proposed guidance scheme for controlling the time of convergence of LOS to its desired values, three different time instants of 10, 15, and 20 s are considered as the convergence time. Simulation results are plotted for these cases in Fig. 2, which depicts trajectories of missile and target, the missile lateral acceleration, the variations of flight path angle of both missile and target, the variation of LOS angle, and its rate.

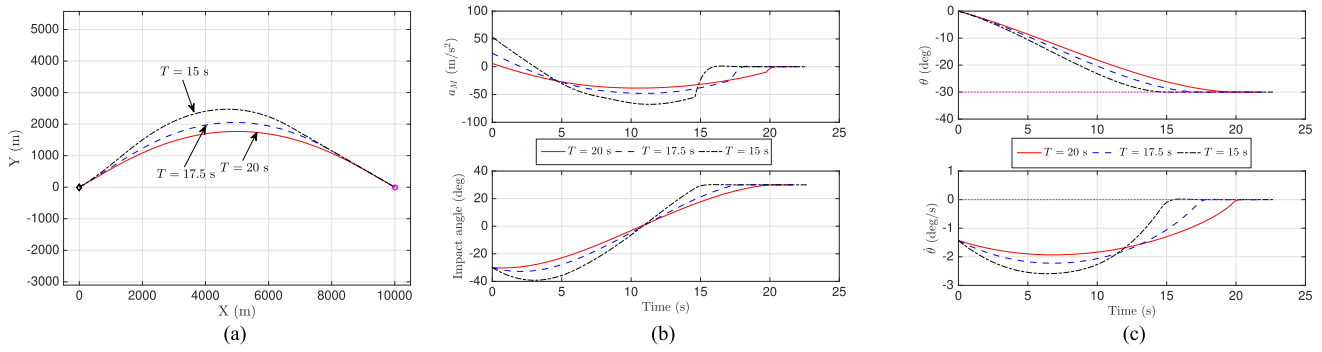


Fig. 2. Stationary target: Interception at different time instants with same launch angle. (a) Trajectories. (b) a_M and $\gamma_T - \gamma_M$. (c) θ and $\dot{\theta}$.

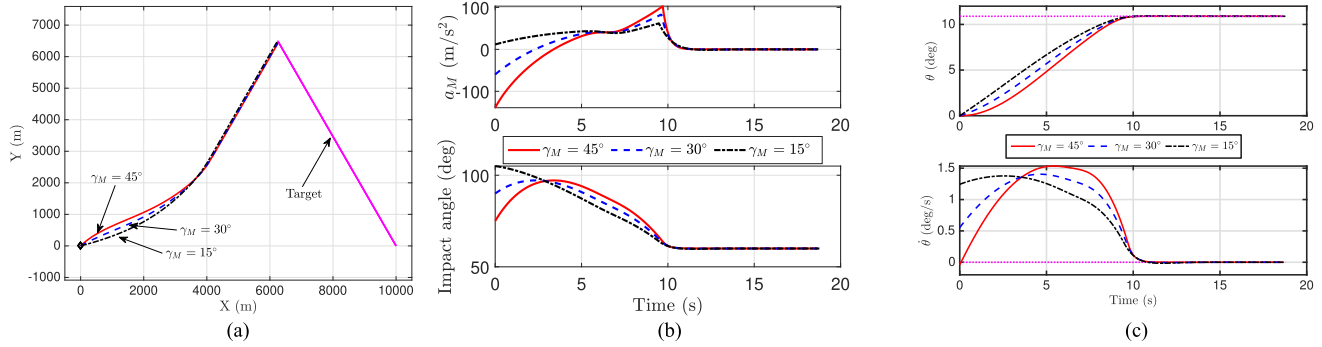


Fig. 3. Constant velocity target: Interception with different launch angles. (a) Trajectories. (b) a_M and $\gamma_T - \gamma_M$. (c) θ and $\dot{\theta}$.

It can be observed from Fig. 2(a) that the missile is able to intercept the target with the proposed guidance strategy. Sliding mode is enforced on the switching surface from the start, and thus the LOS angle and its rate converge to their desired values at specified time instants and thereafter maintain the same, resulting in an interception at desired impact angle of 30° . These observations can be easily obtained from Fig. 2(b) and (c), and it validates the claim of the ability to control convergence time of LOS angle and rate. The profile of missile lateral acceleration is smooth as shown in Fig. 2(b) and it converges to zero after convergence of the LOS angle and its rate to the desired values. The requirement of small acceleration towards interception, as shown in Fig. 2(b), is beneficial.

B. Constant Velocity Targets

In this section, the target is assumed to be of a constant velocity rather than stationary and a desired impact angle of 60° is considered. All initial conditions and simulation parameters remain same as before. However, here the aim is to achieve a convergence time of 10 s. To investigate the performance of proposed guidance scheme under different heading angle errors, three different launch angles of missile, 15° , 30° , and 45° are considered. Simulation results for these cases are plotted in Fig. 3. In these cases also, the missile is able to intercept the target at the desired impact angle, along with convergence of LOS and its rate to their desired values at the specified time. Owing to the relation

between impact angle and LOS angle, even with the different launch angles, the missile is able to engage the target with a desired direction at specified time instant of 10 s. This validates the applicability of the proposed guidance strategy for various launch conditions. Other conclusions similar to Fig. 2 can also be drawn here.

C. Maneuvering Targets

In the previous cases, the target were assumed to be nonmaneuvering. In this case, the maneuvering target with a desired impact angle of 90° is considered. The initial conditions and simulation parameters are the same as in the previous cases, but with an initial LOS angle of 30° . To evaluate performance of the proposed guidance law against target maneuvers, which are considered here as unknown uncertainties, simulations are performed with different levels of target maneuvers. The maneuver of target is assumed to be constant and three different maneuvers of 30, 40, and 50 m/s^2 are considered here. Similar to previous cases, the missile intercepts the target at the desired impact angle, and convergence of LOS and its rates are also achieved in a specified time with all these target maneuvers. This shows robustness against the target maneuvers. Conclusions similar to previous cases can be drawn here. It can be observed from Fig. 4(b) that, unlike the previous cases, the missile lateral accelerations have constant values, after convergence of LOS and its rates, which enforces the missile to move in a circular arc toward interception.

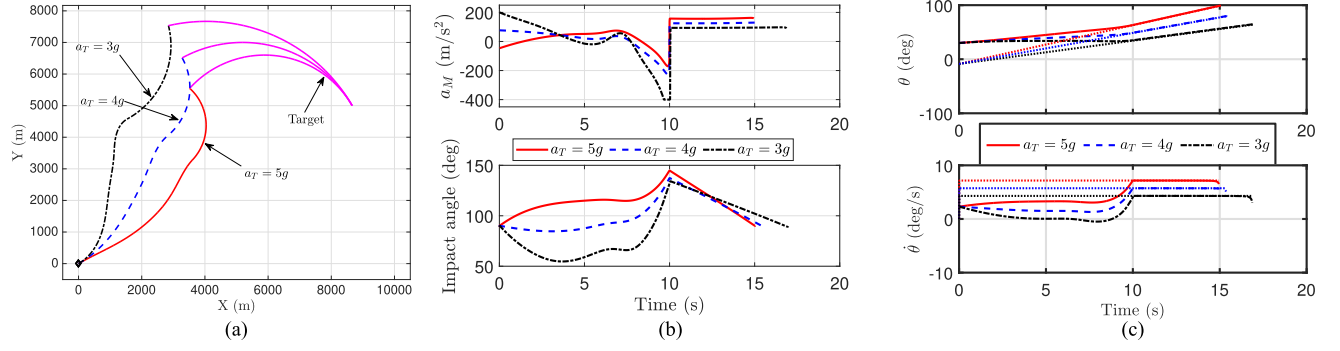


Fig. 4. Maneuvering target: Interception for different target maneuvers. (a) Trajectories. (b) a_M , and $\gamma_T - \gamma_M$. (c) θ and $\dot{\theta}$.

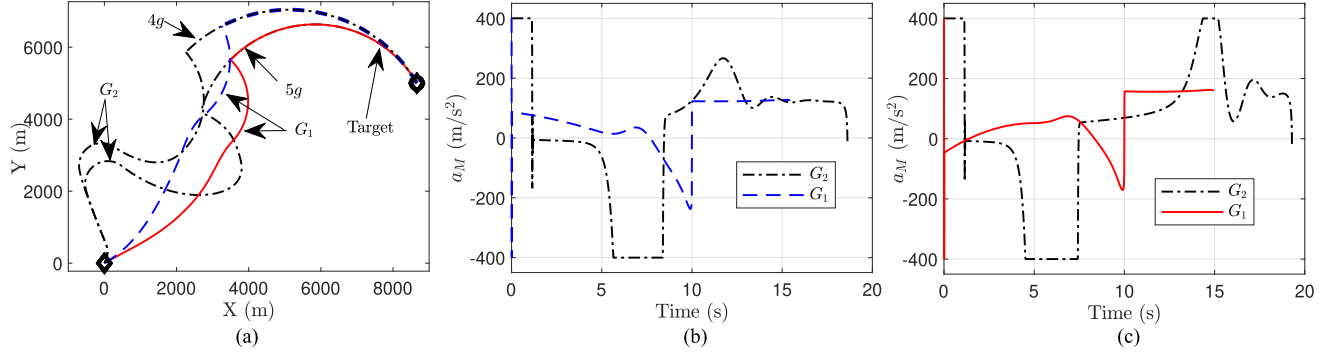


Fig. 5. Comparison of performance of guidance strategies against $a_T = 4g, 5g$. (a) Trajectories with $a_T = 4g, 5g$. (b) a_M for $a_T = 4g$. (c) a_M for $a_T = 5g$.

D. Comparison With Existing Guidance Scheme

To evaluate the relative performance of the proposed guidance strategy, denoted as G_1 , as compared to the existing ones, the performance of proposed strategy is compared with that proposed in [3]. The guidance strategy in [3], denoted as G_2 here, is derived using a nonsingular terminal SMC, which guarantees finite-time convergence to the desired direction. However, the time of convergence is not fixed *a priori*. The simulation results for this comparison is shown in Fig. 5, which depicts the trajectories of missile and target, and their lateral accelerations for the target maneuvers of both $4g$ as well as $5g$. It can be noticed that both the guidance strategies are able to intercept the target successfully. Also, from Fig. 5(b) and (c), it can be observed that the missile lateral acceleration demands with proposed guidance schemes are much lower as compared to that of [3]. Unlike the proposed strategy, the acceleration demand toward interception shoots up for G_2 , which is not desirable. To substantiate the benefit of proposed strategy further, the total integral square control effort, that is, $TCE = \int a_M^2 dt$, is compared for both guidance strategies. The values of TCE for G_1 and G_2 with $a_T = 4g$ are obtained as 1.4746×10^5 and 8.7498×10^5 , respectively, while those with $5g$ are 1.5089×10^5 and 1.0422×10^6 , respectively. These results show the lower demand of total control effort for the proposed guidance strategy. Note that proposed strategy is able to align the missile, at a prespecified time of 10 s, in both cases, which may not be possible with G_2 .

VIII. CONCLUSION

In this paper, a guidance strategy, based on FT-SDRE technique, ensuring alignment of missile in a desired direction at a fixed time was proposed. The proposed FT-SDRE technique ensures that the states of a nonlinear system track their commanded values within a prespecified fixed finite time. This method provides a closed-form solution in terms of solution of the matrix DRE with the specified terminal conditions, which is further expressed as the solutions of algebraic Riccati and Lyapunov equations. This conversion makes the proposed FT-SDRE as computationally efficient, and hence it can be implemented in real time.

The proposed guidance law does not require time-to-go during its implementation, which is a critical issue in most of the optimal control-based guidance strategies. Due to the use of ISM and ST algorithms, proposed strategy was robust throughout the engagement. The performance of proposed strategy was found to be superior than that with an existing guidance strategy. The design of the FT-SDRE technique for a system with bounded control input constraints might be an interesting future direction of research.

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